Tyler J. Roscoe

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Laramie, WY
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Languages: English (native), Mandarin Chinese (elementary proficiency)

Research Interests: Planetary Surface Processes & Climate/Habitability of Terrestrial and Icy Worlds; Applied Geochemistry and Geophysics; Early Earth Systems; Sedimentology & Geomorphology

EDUCATION

Master of Science- University of Wyoming
Expected Graduation Date- May 2028
Major: Geology

Bachelor of Science- University of Wisconsin-Oshkosh
June 2022 - May 2026, GPA 3.8
Major: Geology,
Minors: Physics, Astronomy

RESEARCH PROJECTS

Killpecker Dune Field, Wyoming, as an Analog for Martian Dunes – University of Wyoming Department of Geology and Geophysics
2026-Present

Snowball Earth as an Analog for Enceladus– University of Wisconsin-Oshkosh Department of Geology
2022-Present

Winnebago County Bat Monitoring Project– University of Wisconsin-Oshkosh Department of Environmental Studies and Biology
Summer 2024

X-Ray Diffraction of Single Crystals in Powder XRD Systems- University of Wisconsin-Oshkosh Department of Physics and Astronomy
Spring 2023

IN PREPARATION MANUSCRIPTS

Roscoe, T., and Hiatt, E.E, 2024, Snowball Earth Sub-Ice Biochemical Sediments as Analogs for Europa and Enceladus [Unpublished Manuscript].

EMPLOYMENT

Graduate Research Assistant- University of Wyoming Department of Geology and Geophysics
August 2026-Present

Teaching Assistant- University of Wisconsin-Oshkosh Department of Geology
February 2023-May 2026

Research Assistant- University of Wisconsin-Oshkosh Department of Geology
October 2022-May 2026

Laboratory Technician- University of Wisconsin-Oshkosh Department of Geology
June 2022-May 2026

TECHNICAL SKILLS

- X-Ray Diffraction (XRD) and Analysis
- Brunauer–Emmett–Teller (BET) instrument
- EM31 Ground Conductivity device and Resistivity Mapping
- Ground Penetrating Radar
- Scanning Electron Microscopy (SEM)
- Petrography and Optical Microscopy
- Proton Magnetometer and Magnetic Mapping
- Gravimeter and Gravitational Anomaly Mapping
- Photospectrometry and Surface Water Testing
- Origin Scientific Software
- Scientific and Artistic Illustration
- Microsoft Excel and Office
- ArcGIS Pro and QGIS
- Python coding
- Adobe: Photoshop, Illustrator, After Effects, Premier
- Aquarium and Microecosystem Water Quality and Maintenance (Fresh and Salt water)
- Geologic Mapping

TEACHING APPOINTMENTS

University of Wisconsin-Oshkosh (UWO)

- Lab Teaching Assistant: Geology 101, Geology of the National Parks
- Lab Teaching Assistant: Geology 102, Physical Geology
- Field Teaching Assistant: Environmental Engineering Technology 374/574, Environmental Field Sampling & Analysis

PRESENTATIONS AND CONFERENCES

2026, Geological Society of America Connects 2026

2025, 35th Annual NASA Wisconsin Space Grant Consortium (WSGC) Conference

2025, Wisconsin System (WiSyS), Spark Symposium

2025, UWO Celebration of Scholarship

2024, 34th Annual NASA Wisconsin Space Grant Consortium (WSGC) Conference

2023, Wisconsin System (WiSyS), UWO, “Quick Pitch” Competition

GRANTS RECEIVED

2024, UWO Foundation, C.W. Fetter Endowed Research Fund
Amount Received: $850

2023, NASA Wisconsin Space Grant Consortium Undergraduate Research Scholarship
Amount Received: $4,000

2023, UWO Student/Faculty Collaborative Research Grant
Amount Received: $3,000

AWARDS

2026, UWO Foundation, Burton E. Karges Geology Scholarship
Amount Received: $4,000

2026, UWO Foundation, Burton E. Karges Award for Achievement in Geology
Amount Received: $400

2023, UWO Department of Physics and Astronomy Roy Knispel Physics and Astronomy Scholarship
Amount Received: $750

2023, NASA Wisconsin Space Grant Consortium Undergraduate Scholarship Award
Amount Received: $2,000

2023, Wisconsin System (WiSyS) Research Communication Award
Amount Received: N/A

2023, UWO Geology Scholarship
Amount Received: $850

2022, UWO Geology Scholarship
Amount Received: $850

2022, UWO Ernest O. Thedinga Scholarship

Amount Received: $1,000

ADDITIONAL OUTSIDE COURSES AND WORKSHOPS

Global Energy and Climate Policy
June 2023 University of London, SOAS University of London- through Coursera

Origins- Formations of the Universe, Solar System, Earth, and Life
January 2023 University of Copenhagen- through Coursera

FIELD WORK IN DIVERSE GEOLOGIC SETTINGS

2026: **Wyoming:** Preliminary graduate field work in Killpecker Dune Field,

Geology Field Camp Course (UWO),

Graduate "Rocky Mountain Field Trip" University of Wyoming

Eastern Nevada: field mapping project for Geology Field Camp Course (UWO)

Utah (Various): Various field mapping and sampling projects for Geology Field Camp Course (UWO)

South Dakota: Black Hills: field mapping project for Geology Field Camp Course (UWO)

Upper Peninsula of Michigan: Structural Geology Class project

2025: **Florida Keys:** Biological sampling of Diatoms

Wisconsin: Assisted teaching field trip in Michigan Basin,

Monitored Bat populations on Lake Winnebago

Northern Illinois: examined and sampled Fossils within Bond and Mattoon Formation

2024: **Wisconsin:** Assisted teaching in field course in Door Peninsula,

Monitored Bat populations on Lake Winnebago

Northern Illinois: examined and sampled Fossils within Bond and Mattoon Formation

2023: **Wisconsin:** Assisted teaching in course field trip to Kettle Moraine glacial terrains,

Conducted biological survey of lichen populations in eastern Wisconsin

Bermuda: Biological and geological, terrestrial and oceanic field sampling as part of a field course at Bermuda Institute of Oceanographic Sciences (BIOS)

2022: **Wisconsin:** Sampling in Michigan Basin and Coast of Lake Michigan

MEMBERSHIPS IN PROFESSIONAL SOCIETIES AND OUTREACH EFFORTS

Geological Society of America
2026 Member
UWO Undergraduate Research and Creative Activities Club
2025-2026 Member
UWO Community Free School
Spring 2025 Volunteer Instructor
UWO Geology Club
2022-2026 Member
Society of Physics Students
2022-2026 Member, Vice President of UWO Chapter (Spring 2025)
American Physical Society
2022-2023 Member
University of Wisconsin Green Bay Lower Fox River Monitoring Program
2021-2022 Volunteer

CAPSTONE ACADEMIC PROJECTS

2026, M11 "Wild Duck Cluster" Star Cluster: Zero Age Main Sequence, and Metalicity

2025, The Origin and Consequences of our Solar System, Earth, and Life; Term Research Paper

2025, Gravitational Anomaly Mapping Project of Winnebago County, WI

2023, Biological Survey of Lichen Populations in Wisconsin

2023, Geomorphology of the Dwarf Planet Ceres Term Paper and Presentation